

Engineering Handbook – Volume 05

Backend Architecture Handbook (Version 1.0)

Objective

Describe the backend architecture, project organization and engineering conventions.

Framework

FastAPI, SQLAlchemy ORM, PostgreSQL, Pydantic and JWT form the backend foundation.

Project Structure

Organize code into routers, models, schemas, database, security and future services.

Startup Flow

Initialize configuration, database engine, routers and OpenAPI documentation during application startup.

Database Session

Each request receives an isolated SQLAlchemy session. Commit on success and rollback on exceptions.

Router Responsibilities

Routers should validate requests and delegate business logic to service classes.

Schema Responsibilities

Schemas define API contracts and isolate external interfaces from ORM models.

Authentication

JWT authentication with bcrypt password hashing protects secured endpoints.

Error Handling

Return consistent JSON errors, log unexpected failures and avoid exposing internals.

Configuration

Read secrets and connection strings from environment variables.

Scalability

Future modules include Redis, Celery, MQTT, notifications and analytics services.

Engineering Rules

Keep modules cohesive, avoid circular imports, maintain backward compatibility and document changes.

Future Backend Modules

Service Layer, Repository Layer, MQTT Consumer, Telemetry Processor, Alarm Engine, Notification Service, Reporting Service, Audit Logging and Device Provisioning.